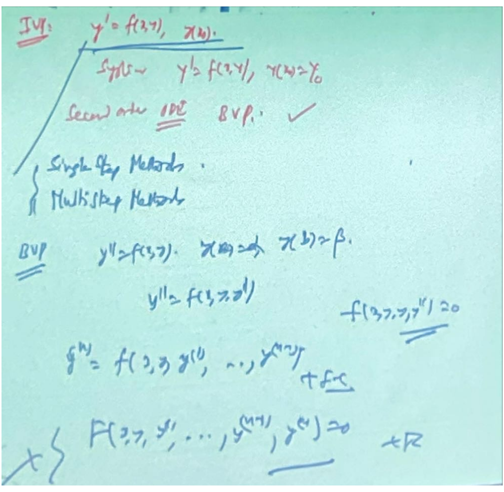
Introduction

Friday, 25 July 2025 9:51 AM



IVP: $y' = f(x,y)$, $x(x_0)$.

Syst~ $y' = f(x,y)$, $y(x_0) = y_0$

Second order ODE BVP. ✓

Single Step Methods.

Multistep Methods

BVP $y'' = f(x,y)$. $x(a) = \alpha$, $x(b) = \beta$.

$y'' = f(x,y,y')$

$f(x,y,y',y'') = 0$

$y^{(n)} = f(x,y,y^{(1)},\dots,y^{(n-1)})$ + f.c.

$F(x,y,y',\dots,y^{(n-1)},y^{(n)}) = 0$ + R

PDE (2nd order) → Parabolic $\dfrac{\partial u}{\partial y} + \alpha \dfrac{\partial^2 u}{\partial x^2} = 0$

→ Hyperbolic $\dfrac{\partial^2 u}{\partial y \partial x} + \alpha \dfrac{\partial^2 u}{\partial x^2} = 0$

→ Elliptic (2D) $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

3D elliptic → $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$

☆ $y' = f(x,y)$ $y(x_0) = y_0$ Given this eq. and the value at a point x_0, we want to find solution to a set of points $x_0 = a, \dots\dots x_n = b$ in the range $[a,b]$

These points are called abscissae / nodal points and they are equally spaced. $x_{k+1} - x_k = h = \dfrac{b-a}{\underset{\hookrightarrow \text{no. of intervals}}{n}}$

Objective To find numerically, the solution at these n+1 nodal points.

Numerical solution vs. analytical / closed form solution

- $y' = f(x,y)$, $y(x_0) = y_0 \iff y(x) = y_0 + \displaystyle\int_{x_0}^{x} f(t, y(t))\,dt$
- General problems encountered may not have a method to get to the analytical solution, but numerical solutions can always be obtained.

Picard's iteration method → $y_{n+1} = \underset{(y0)}{y_n(x_0)} + \displaystyle\int_{x_0}^{x} f(s, y_n(s))\,ds$

(start with $y_0(x) = y_0$)

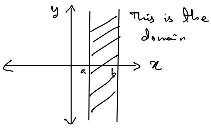
Methods to solve → Single step methods - To find the solution at x_{n+1}, we need the solution at x_n.

→ Multistep methods - To find the solution at x_{n+1}, we need the solution of more than one previous nodal point (atleast at x_n and x_{n-1})

Eg. - We need solution at $x_n, x_{n-1}, \dots x_{n-k+1}$ (k points).

∴ We call this k-step method.

A unique solution of $y' = f(x,y)$, $y(x_0) = y_0$ exists in $[a,b]$ when $f(x,y)$ is continuous and bounded in the domain $a = x_0 \leq x \leq b$, $-\infty < y < \infty$.



This is the domain

i.e., ∃ a constant L (Lipschitz constant) s.t. for $x \in [a,b]$ and y, y^*

$$|f(x,y) - f(x,y^*)| \leq L|y - y^*|$$

Single step methods

☆ Taylor series method / Taylor method

Suppose y at x_n, i.e., $y(x_n)$ is known.

Objective → To find y at x_{n+1} (the next nodal point)

$y(x_{n+1}) = y(x_n + h)$. Expanding by Taylor series (assume that the derivative upto the order of expansion is well-defined)

$$y(x_n + h) = y(x_n) + h y'(x_n) + \frac{h^2}{2!} y''(x_n) \dots + \frac{h^p}{p!} y^{(p)}(x_n) + \underbrace{\frac{h^{p+1}}{(p+1)!} y^{(p+1)}(\xi)}_{\text{LTE}}$$

We have $y' = f(x,y)$. Neglecting the last term / error term, we get

$$y(x_{n+1}) \simeq y(x_n) + h y'(x_n) + \frac{h^2}{2!} y''(x_n) + \dots + \frac{h^p}{p!} y^{(p)}(x_n)$$

- The neglected term is called local truncation error (LTE)

Denoting the approx. for $y(x_n)$ by y_n, we get

$$y(x_{n+1}) = y_{n+1} \simeq y_n + h y'_n + \frac{h^2}{2!} y''_n + \dots + \frac{h^p}{p!} y^{(p)}_n$$

① $\boxed{y_{n+1} = y_n + h y_n' + \dots + \dfrac{h^p}{p!} y_n^{(p)}}$ ⇒ p^{th} order Taylor series method

Note that $y(x_0) = y_0$ is given to us initially. $y_n = y(x_n)$, $n \geq 1$

$y(x_{n+1}) - \underset{\hookrightarrow \text{given by ①}}{y_{n+1}}$, is called the total truncation error, and it is denoted by e_{n+1}.

$$e_{n+1} = y(x_{n+1}) - y_{n+1}$$

If $p = 1$, this method is called Euler's method.

$\boxed{y_{n+1} = y_n + h y_n'}$ Here, $y_n' = f(x_n, y_n)$

$$\Rightarrow \quad y_{n+1} = y_n + h f(x_n, y_n)$$

Putting $n = 0, 1, 2, \dots$

$y_1 = y_0 + h f(x_0, y_0)$

$y_2 = y_1 + h f(x_1, y_1)$

$y_3 = y_2 + h f(x_2, y_2)$

☆ Successive derivatives → $y_n', y_n'', y_n''', \ldots$ How to compute these?

Suppose $y' = f(x, y)$

$$y'' = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\boxed{\frac{\partial y}{\partial x}} = f_x + f f_y$$

$\frac{\partial y}{\partial x} \rightarrow f$

$$y''' = \left[\frac{\partial f_x}{\partial x} + \frac{\partial f_x}{\partial y}\cdot\frac{\partial y}{\partial x}\right] + \left[\frac{\partial}{\partial x}(f f_y) + \frac{\partial}{\partial y}(f f_y)\frac{\partial y}{\partial x}\right]$$

$$= f_{xx} + f f_{xy} + f_x f_y + f f_{xy} + f(f f_{yy} + f_y f_y)$$

$$= f_{xx} + 2 f f_{xy} + f f f_{yy} + f_y (f_x + f f_y)$$

Problem Find solution of IVP $y' = -xy$ $y(0) = 1$ at $x = 0.1$ and 0.2 using Taylor's second order method.

→ Can also phrase as compute 2 iterations with step size 0.1

Solution

$$y_{n+1} = y_n + h y_n' + \frac{h^2}{2!} y_n''$$

$x_0 = 0$, $y_0 = 1$ $x_1 = 0.1$, $x_2 = 0.2$ → $h = 0.1$

$n = 0$ → $y_1 = y_0 + h y_0' + \frac{h^2}{2!} y_0''$

$y' = f(x, y) = -xy$

$y_0' = -x_0 y_0 = 0$

$y'' = -y - xy'$

$= -y - x(-xy)$

$= -y + x^2 y$

$y_0'' = -y_0 + x_0^2 y_0 = -1$

$y_1 = 1 + 0 + \frac{0.1^2}{2}(-1)$

$= 1 - \frac{0.01}{2}$

$= 1 - 0.005$

$= 0.9950$

(maintain 4 decimal places in all answers)

$y_2 = y_1 + h y_1' + \frac{h^2}{2!} y_1''$

$y_1' = -x_1 y_1 = -0.1 \times 0.995 = -0.0995$

$y_1'' = -y_1 - x_1^2 y_1 = -0.995 - 0.1^2 \times 0.995 = -0.9851$

$y_2 = 0.995 - 0.00995 + \frac{0.1^2}{2} \times (-0.9851) = 0.9801$

Convergence of a numerical method

Let y_n be an approximation to the exact value $y(x_n)$ at x_n where $x_n = x_0 + nh$. If $\lim_{n\to\infty} y_n = y(x_n)$ then the method by which y_n is computed is said to be convergent.

$$\boxed{\lim_{n\to\infty} (y(x_n) - y_n) \longrightarrow 0 \quad \text{, i.e., } \lim_{n\to\infty} e_n = 0}$$

Lemma If $w_0, w_1 \ldots$ is a seq. of real no.s s.t.

$w_{k+1} \le (1+a) w_k + B$, where a, B are +ve no.s, then

$$w_n \le \frac{B(e^{an} - 1)}{a}, \text{ where } w_0 = 0$$

Proof (By induction)

Since $w_0 = 0$, the ineq. $w_0 = 0 \le \frac{B(e^0 - 1)}{a}$ is satisfied by w_0.

Assume that the result is true for $n = k-1$. Now prove that it is true for $n = k$.

$$w_{k-1} \le \frac{B(e^{a(k-1)} - 1)}{a}$$

We also have $w_k \le (1+a) w_{k-1} + B$

$$\le (1+a) B \left(\frac{e^{a(k-1)} - 1}{a}\right) + B$$

$$= B\left[(1+a)\left(\frac{e^{a(k-1)} - 1}{a}\right) + 1\right]$$

$$= B\left[\frac{(1+a)e^{a(k-1)} - 1 - \cancel{a} + \cancel{a}}{a}\right]$$

$$= B\left[\frac{(1+a)e^{a(k-1)} - 1}{a}\right]$$

Since $a > 0$, $1 + a \le e^a$

$$\Rightarrow w_k \le \frac{B}{a}\left[e^a \cdot e^{a(k-1)} - 1\right] = \frac{B(e^{ak} - 1)}{a}$$

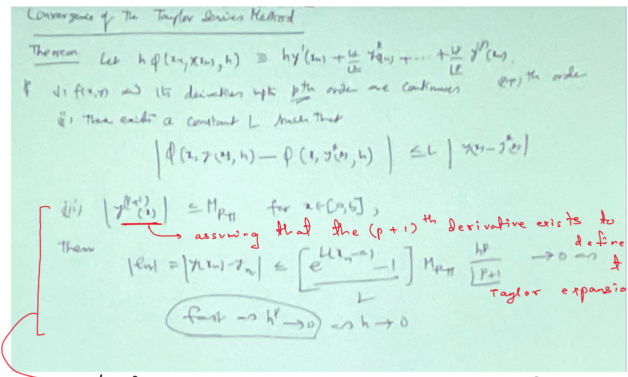
Hence proved.

Convergence of Taylor series method

Theorem Let $h\phi(x_n, y(x_n), h) = hy'(x_n) + \frac{h^2}{2!} y''(x_n) + \ldots + \frac{h^p}{p!} y^{(p)}(x_n)$

If $f(x, y)$ and its derivatives upto p^{th} order are cont., ∃ a constant L s.t.

$$|\phi(x, y(x), h) - \phi(x, y^*(x), h)| \le L|y(x) - y^*(x)|$$



Convergence of The Taylor Series Method

Theorem Let $h\phi(x_n, y(x_n), h) \equiv hy'(x_n) + \frac{h^2}{2!} y''(x_n) + \cdots + \frac{h^p}{p!} y^{(p)}(x_n)$.

If (i) $f(x,y)$ and its derivatives upto pth order are continuous $(p+1)$th order

(ii) There exists a constant L such that

$$|\phi(x, y(x), h) - \phi(x, y^*(x), h)| \le L|y(x) - y^*(x)|$$

(iii) $|y^{(p+1)}(x)| \le M_{p+1}$ for $x \in [a,b]$,

Then

$$|e_n| = |y(x_n) - y_n| \le \left[\frac{e^{L(x_n - a)} - 1}{L}\right] M_{p+1} \frac{h^p}{(p+1)!} \to 0 \text{ as}$$

fast → $h^p \to 0$ as $h \to 0$

→ assuming that the $(p+1)^{th}$ derivative exists to be able to define the error term of the Taylor expansion.

$$y^{(p+1)}(x) \le M_{p+1} \quad \forall x \in [a, b]$$

Then $|e_n| = |y(x_n) - y_n| \le \left[\frac{e^{L(x_n - a)} - 1}{L}\right] M_{p+1} \frac{h^p}{(p+1)!} \to 0$

as fast as $h^p \longrightarrow 0$ as $h \longrightarrow 0$.

Proof We have $y(x_{n+1}) = y(x_n) + h\phi(x_n, y(x_n), h) + \frac{h^{p+1}}{(p+1)!} y^{(p+1)}(\xi)$ – ①

$(x_n < \xi < x_{n+1})$

[Actual value by p^{th} order Taylor expansion around x_n]

p^{th} order Taylor series method ⇒

$$y_{n+1} = y_n + h y_n' + \frac{h^2}{2!} y_n'' + \ldots + \frac{h^p}{p!} y^{(p)} \quad \text{– ②}$$

① – ② ⇒

$$y(x_{n+1}) - y_{n+1} = y(x_n) - y_n + h\phi(x_n, y(x_n), h) - h\phi(x_n, y_n, h) + \frac{h^{p+1}}{(p+1)!} y^{(p+1)}(\varepsilon)$$

$$|y(x_{n+1}) - y_{n+1}| \le |y(x_n) - y_n| + h\,|\phi(x_n, y(x_n), h) - \phi(x_n, y_n, h)| + \frac{h^{p+1}}{(p+1)!} |y^{(p+1)}(\varepsilon)| \qquad \le M_{p+1}$$

But by hypothesis,

$$|\phi(x, y(x_n), h) - \phi(x, y_n, h)| \le L\,|y(x_n) - y_n|$$

$$|y(x_{n+1}) - y_{n+1}| \le |y(x_n) - y_n| + hL\,|y(x_n) - y_n| + \frac{h^{p+1}}{(p+1)!} M_{p+1}$$

$$|e_{n+1}| \le (1 + hL)\,|y(x_n) - y_n| + \frac{h^{p+1}}{(p+1)} M_{p+1}$$

$$\boxed{|e_{n+1}| \le (1 + hL)\,|e_n| + \frac{h^{p+1}}{(p+1)!} M_{p+1}} \quad - ③$$

Now, applying lemma proved previously, as the seq. of no.s $|e_0|, |e_1|, \dots$ satisfies the ineq., it follows that

$$|e_n| \le \frac{h^{\cancel{p+1}}}{(p+1)!} M_{p+1} \frac{\left(e^{nhL} - 1\right)}{\cancel{h}L} \qquad [nh = x_n - x_0]$$

$$|e_n| \le \frac{h^{p}}{(p+1)!} M_{p+1} \frac{\left(e^{(x_n - x_0)L} - 1\right)}{L}$$

$$|e_n| \le \frac{h^{p}}{(p+1)!} M_{p+1} \frac{\left(e^{(x_n - a)L} - 1\right)}{L} \longrightarrow 0$$

as $h \longrightarrow 0$. $\therefore e_n \longrightarrow 0$ as $h \to 0$.

Hence, the method is convergent.

If the local truncation error associated with a numerical method is of order h^{p+1}, then the method is called p^{th} order method.

- MVT (mean value theorem of diff. eq.) → f is cont. on $[a, b]$ & diff. on (a, b). Then $\exists$ a const. $\varepsilon \in (a, b)$ s.t.

$$f(b) - f(a) = f'(\varepsilon)(b - a)$$

In case of DE $y' = f(x, y)$, if $y(x)$ is solution, then we write

$$y(x_{n+1}) - y(x_n) = (x_{n+1} - x_n)\, y'(\varepsilon)$$

If we take x_n, x_{n+1} (consecutive nodal points), then $x_{n+1} - x_n = h$. $\therefore y(x_{n+1}) = y(x_n) + hy'(\varepsilon)$

$$\boxed{y(x_{n+1}) = y(x_n) + hf(\varepsilon, y(\varepsilon))}$$

All that we know is $y(x_0) = y_0$.
Approximation of $y(x_{n+1})$ can be done by approximating the value of $f(\varepsilon, y(\varepsilon))$

Approx. I

$$f(\varepsilon, y(\varepsilon)) = \frac{1}{2}\left[f(x_n, y(x_n)) + f(x_{n+1}, y(x_{n+1}))\right] - ①$$

$$y(x_{n+1}) = y(x_n) + \frac{h}{2}\left[f(x_n, y(x_n)) + f(x_{n+1}, \underbrace{y(x_{n+1})}_{\text{To be computed}})\right]$$

obtain an approx. value of this for the RHS using Euler method.

$$y_{n+1} = y_n + hf(x_n, y_n)$$

Substituting this value in eq. ①, we get ⇒

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_n + hf(x_n, y_n))\right]$$

Using the approximations in ①, we get ⇒

$$\boxed{\begin{aligned} y_{n+1} &= y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right] - ② \\ \text{where } y_{n+1} &= y_n + hf(x_n, y_n) - ③ \text{ on RHS.} \end{aligned}}$$

② is called improved Euler method / trapezoidal / trapezium method. This is a special case of 2nd order Runge-Kutta method. It is also called Heun's method.

It is called trapezoid method because $\int_{x_n}^{x_{n+1}} y'\,dx = \int_{x_n}^{x_{n+1}} f(x, y(x))\,dx$

$$\simeq \frac{1}{2} h\left[f(x_n, y(x_n)) + f(x_{n+1}, y(x_{n+1}))\right]$$

(approximation by the area of the trapezium under $f(x, y(x))$)

Define $\boxed{k_1 = hf(x_n, y_n), \quad k_2 = hf(x_n + h, y_n + k_1)}$ $- ④$

Now, ② becomes

$$y_{n+1} = y_n + \frac{1}{2}\left[hf(x_n, y_n) + hf(x_n + h, y_n + k_1)\right]$$

By substituting ④ ⇒

$$\boxed{y_{n+1} = y_n + \frac{1}{2}\left[k_1 + k_2\right]} \quad - ⑤$$

④ and ⑤ together are called 2nd order Runge Kutta method.

∴ The improved euler method can be represented as 2nd order R-K method as shown above in ④ and ⑤.

Approx. II

From MVT, we have

$$y(x_{n+1}) = y(x_n) + h(\varepsilon, f(\varepsilon))$$

Suppose we ε to be the midpoint of x_n and x_{n+1}, i.e.,

$$\varepsilon = x_n + \frac{h}{2}.$$

Then, $\boxed{y(x_{n+1}) = y(x_n) + hf\left(x_n + \frac{h}{2}, y\left(x_n + \frac{h}{2}\right)\right)}$ $- ①$

$y\left(x_n + \frac{h}{2}\right)$ is calculated using Euler's method, i.e.,

$$\boxed{y\left(x_n + \frac{h}{2}\right) \simeq y_n + \frac{h}{2} f(x_n, y_n)} \quad - ②$$

From ① and ②, we get approximation y_{n+1} for $y(x_{n+1})$ by the following ⇒

$$\boxed{y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right)} \quad - ③$$

This is called modified Euler method. Now, put $k_1 = hf(x_n, y_n)$ and $k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$

$\therefore y_{n+1} = y_n + k_2$. This is a special case of 2nd order R-K method.

★ $f_n = f(x,y)\Big|_{(x_n,y_n)}$

General 2nd order R-K method → We define

$$\boxed{\begin{aligned} k_1 &= hf(x_n, y_n) = hy_n' = hf_n \\ k_2 &= hf(x_n + \alpha_2 h,\ y_n + \beta_{21} k_1) \\ y_{n+1} &= y_n + w_1 k_1 + w_2 k_2, \\ &\text{where } \alpha_2, \beta_{21}, w_1, w_2 \text{ are constants to be determined.} \end{aligned}}$$

Expanding k_2 in Taylor series about (x_n, y_n), we get

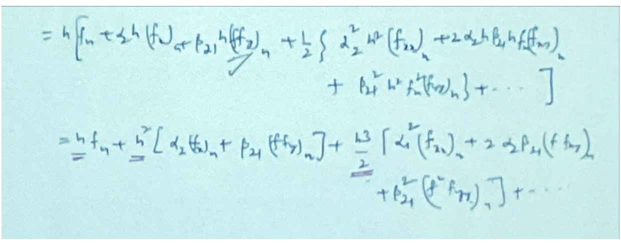
$$k_2 = h\Big[f(x_n,y_n) + \left(\alpha_2 h \frac{\partial}{\partial x} + \beta_{21} k_1 \frac{\partial}{\partial y}\right) f_n + \frac{1}{2!}\left(\alpha_2 h \frac{\partial}{\partial x} + \beta_{21} k_1 \frac{\partial}{\partial y}\right)^2 f_n + \dots\Big]$$

$$= h\Big[f_n + \alpha_2 h (f_x)_n + \beta_{21} k_1 (f_y)_n + \frac{1}{2!}\left(\alpha_2^2 h^2 \frac{\partial^2}{\partial x^2} + 2\alpha_2 \beta_{21} h k_1 \frac{\partial^2}{\partial x \partial y} + \beta_{21}^2 k_1^2 \frac{\partial^2}{\partial y^2}\right) f_n + \dots\Big]$$

$$= h\Big[f_n + \alpha_2 h (f_x)_n + \beta_{21} k_1 (f_y)_n + \frac{1}{2!}\left(\alpha_2^2 h^2 (f_{xx})_n + 2\alpha_2 \beta_{21} h k_1 (f_{xy})_n + \beta_{21}^2 k_1^2 (f_{yy})_n\right) + \dots\Big]$$

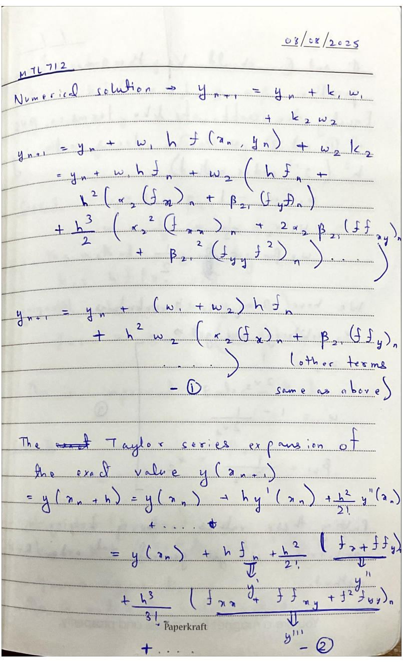
$$\left[(f_x)_n = \frac{\partial f}{\partial x}\Big|_{(x_n,y_n)} \qquad f_{xx} = \frac{\partial^2 f}{\partial x^2}\Big|_{(x_n,y_n)}\right]$$

Put $k_1 = hf_n$.



$$= h\Big[f_n + \alpha_2 h (f_x)_n + \beta_{21} h (f f_y)_n + \frac{1}{2}\left\{\alpha_2^2 h^2 (f_{xx})_n + 2\alpha_2 h \beta_{21} h f_n (f_{xy})_n + \beta_{21}^2 h^2 f_n^2 (f_{yy})_n\right\} + \dots\Big]$$

$$= hf_n + h^2\left[\alpha_2 (f_x)_n + \beta_{21} (f f_y)_n\right] + \frac{h^3}{2}\left[\alpha_1^2 (f_{xx})_n + 2\alpha_2 \beta_{21} (f f_{xy})_n + \beta_{21}^2 (f^2 f_{yy})_n\right] + \dots$$



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Numerical solution → $y_{n+1} = y_n + k_1 w_1 + k_2 w_2$

$$y_{n+1} = y_n + w_1 h f(x_n, y_n) + w_2 k_2$$

$$= y_n + w_1 h f_n + w_2 \Big(h f_n + h^2\left(\alpha_2 (f_x)_n + \beta_{21} (f_y f)_n\right) + \frac{h^3}{2}\left(\alpha_2^2 (f_{xx})_n + 2\alpha_2 \beta_{21} (f f_{xy})_n + \beta_{21}^2 (f_{yy} f^2)_n\right)\dots\Big)$$

$$y_{n+1} = y_n + (w_1 + w_2) h f_n + h^2 w_2 \left(\alpha_2 (f_x)_n + \beta_{21} (f f_y)_n \dots\right) \quad \text{— ①}$$

(other terms same as above)

The ~~exact~~ Taylor series expansion of the exact value $y(x_{n+1})$

$$= y(x_n + h) = y(x_n) + h y'(x_n) + \frac{h^2}{2!} y''(x_n) + \dots$$

$$= y(x_n) + h \underbrace{f_n}_{y'} + \frac{h^2}{2!} \underbrace{(f_x + f f_y)}_{y''} + \frac{h^3}{3!} \underbrace{(f_{xx} + f f_{xy} + f^2 f_{yy})_n}_{y'''} + \dots \quad \text{— ②}$$

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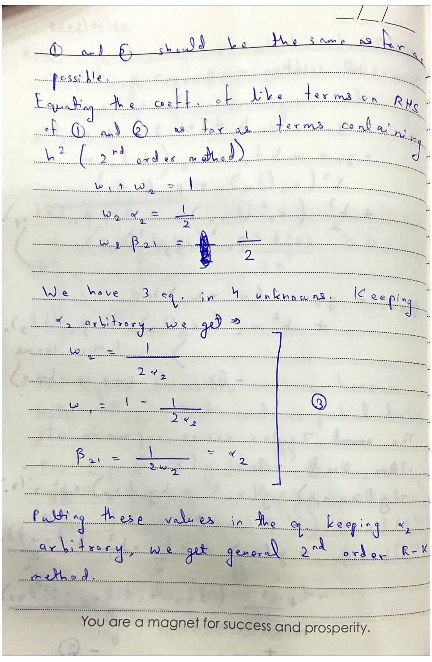

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① and ② should be the same as far as possible.

Equating the coeff. of like terms on RHS of ① and ② as far as terms containing h^2 (2nd order method)

$$w_1 + w_2 = 1$$

$$w_2 \alpha_2 = \frac{1}{2}$$

$$w_2 \beta_{21} = \frac{1}{2}$$

We have 3 eq. in 4 unknowns. Keeping α_2 arbitrary, we get ⇒

$$\left.\begin{aligned} w_2 &= \frac{1}{2\alpha_2} \\ w_1 &= 1 - \frac{1}{2\alpha_2} \\ \beta_{21} &= \frac{1}{2w_2} = \alpha_2 \end{aligned}\right] \quad ③$$

Putting these values in the eq. keeping α_2 arbitrary, we get general 2nd order R-K method.

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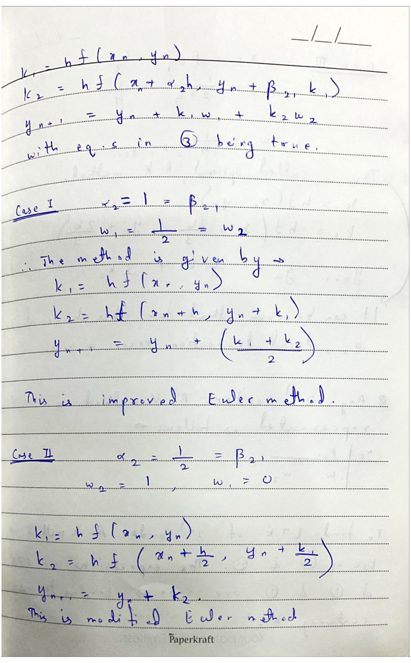

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$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + \alpha_2 h,\ y_n + \beta_{21} k_1)$$

$$y_{n+1} = y_n + k_1 w_1 + k_2 w_2$$

with eq.s in ③ being true.

Case I $\qquad \alpha_2 = 1 = \beta_{21}$

$$w_1 = \frac{1}{2} = w_2$$

∴ The method is given by →

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h,\ y_n + k_1)$$

$$y_{n+1} = y_n + \left(\frac{k_1 + k_2}{2}\right)$$

This is improved Euler method.

Case II $\qquad \alpha_2 = \frac{1}{2} = \beta_{21}$

$$w_2 = 1, \quad w_1 = 0$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2},\ y_n + \frac{k_1}{2}\right)$$

$$y_{n+1} = y_n + k_2.$$

This is modified Euler method

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Case III $\alpha_2 = \beta_{21} = \frac{2}{3}$

$$w_2 = \frac{1}{2 \times \frac{2}{3}} = \frac{3}{4}$$

$$w_1 = \frac{1}{4}$$

This is called optimal 2nd order R-K method.

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{2h}{3}, y_n + \frac{2k_1}{3}\right)$$

$$y_{n+1} = y_n + \frac{1}{4}k_1 + \frac{3}{4}k_2$$

It can be seen that for this choice of α_2, local truncation error is min.

* Any 2nd order R-K method is represented as follows →

$$\begin{array}{c|cc} \alpha_2 & \beta_{21} & \\ \hline & w_1 & w_2 \end{array}$$

To find LTE of 2nd order R-K method →

Use eq. ① & ②. The expressions on RHS of ① & ② are same as far as terms containing h^2.

Your potential is limitless.

$$y(x_{n+1}) - y_{n+1}$$

$$\text{LTE} = \frac{h^3}{6} y_n''' - w_2 \left(\alpha_2^2 f_{xx} + 2\alpha_2 \beta_{21} f f_{xy} + \beta_{21}^2 f^2 f_{yy}\right)_n \frac{h^3}{2}$$

+ terms containing higher powers of h.

Now, use $\beta_{21} = \alpha_2$

$$\text{LTE} = \frac{h^3}{6} y_n''' - w_2 \alpha_2^2 \left(f_{xx} + 2f f_{xy} + f^2 f_{yy}\right)_n \frac{h^3}{2} + \dots$$

Now, use $w_2 = \frac{1}{2\alpha_2}$

$$\text{LTE} = \frac{h^3}{6} y_n''' - \frac{1}{2} \cdot \alpha_2 \cdot \frac{h^3}{2} \left(f_{xx} + 2f f_{xy} + f^2 f_{yy}\right)_n + \dots$$

$$= \frac{h^3}{6} y_n''' - \frac{h^3}{4} \alpha_2 \left(y_n''' - (f_y)_n y_n''\right)$$

$$\star\ y_n''' = \left(f_{xx} + 2ff_{xy} + f^2 f_{yy}\right)_n + (f_y)_n \underbrace{(f_x + ff_{yy})_n}_{y_n''}$$

$$\text{LTE} = \frac{h^3}{12}\left[2y_n''' - 3\alpha_2\left[y_n''' - (f_y)_n y_n''\right]\right] + \dots$$

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$$= \frac{h^3}{12}\left[(2 - 3\alpha_2) y_n''' + 3\alpha_2 (f_y)_n y_n''\right] + \dots$$

The principal error term of the first order RK method is $\frac{h^3}{12}\left[(2 - 3\alpha_2) y_n''' + 3\alpha_2 (f_y)_n y_n''\right]$, where α_2 is arbitrary.

When $\alpha_2 = \frac{2}{3}$, the 1st term becomes 0, making the error constant small. ∴ It is numerically least when $\alpha_2 = 2/3$ & the corresponding method is called optimal 2nd order RK method.

Higher order RK methods

3rd order RK method ⇒

$$\begin{array}{c|ccc} \alpha_2 & \beta_{21} & & \\ \alpha_3 & \beta_{31} & \beta_{32} & \\ \hline & w_1 & w_2 & w_3 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + \alpha_2 h, y_n + \beta_{21} k_1)$$

$$k_3 = hf(x_n + \alpha_3 h, y_n + \beta_{31} k_1 + \beta_{32} k_2)$$

$$y_{n+1} = y_n + w_1 k_1 + w_2 k_2 + w_3 k_3$$

(8 unknowns)

Some 3rd order RK methods →

①

$$\begin{array}{c|ccc} 1/3 & 1/3 & & \\ 1/3 & 0 & 2/3 & \\ \hline & 1/4 & 0 & 3/4 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h/3, y_n + k_1/3)$$

$$k_3 = hf(x_n + 2h/3, y_n + 2k_2/3)$$

$$y_{n+1} = y_n + \frac{k_1}{4} + \frac{3k_3}{4}$$

This method is called Heun's method.

② 3rd order classical method

$$\begin{array}{c|ccc} 1/2 & 1/2 & & \\ 1 & -1 & 2 & \\ \hline & 1/6 & 2/3 & 1/6 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf(x_n + h, y_n - k_1 + 2k_2)$$

$$y_{n+1} = y_n + \frac{k_1}{6} + \frac{2k_2}{3} + \frac{k_3}{6}$$

③ Nystraum's formula

$$\begin{array}{c|ccc} 2/3 & 2/3 & & \\ 2/3 & 0 & 2/3 & \\ \hline & 2/8 & 3/8 & 3/8 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{2h}{3}, y_n + \frac{2k_1}{3}\right)$$

$$k_3 = hf\left(x_n + \frac{2h}{3}, y_n + \frac{2k_2}{3}\right)$$

$$y_{n+1} = y_n + \frac{2k_1}{8} + \frac{3k_2}{8} + \frac{3k_3}{8}$$

④ optimal

$$\begin{array}{c|ccc} 1/2 & 1/2 & & \\ 3/4 & 0 & 3/4 & \\ \hline & 2/9 & 3/9 & 4/9 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{3h}{4}, y_n + \frac{3k_2}{4}\right)$$

$$y_{n+1} = y_n + \frac{1}{9}(2k_1 + 3k_2 + 4k_3)$$

4th order RK method (explicit method)

$$\begin{array}{c|cccc} \alpha_2 & \beta_{21} & & & \\ \alpha_3 & \beta_{31} & \beta_{32} & & \\ \alpha_4 & \beta_{41} & \beta_{42} & \beta_{43} & \\ \hline & w_1 & w_2 & w_3 & w_4 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + \alpha_2 h, y_n + \beta_{21} k_1)$$

$$k_2 = hf(x_n + \alpha_3 h, y_n + \beta_{31} k_1 + \beta_{32} k_2)$$

$$k_3 = hf(x_n + \alpha_4 h, y_n + \beta_{41} k_1 + \beta_{42} k_2 + \beta_{43} k_3)$$

$$y_{n+1} = y_n + w_1 k_1 + w_2 k_2 + w_3 k_3 + w_4 k_4$$

** Classical 4th order RK method

$$\begin{array}{c|cccc} 1/2 & 1/2 & & & \\ 1/2 & 0 & 1/2 & & \\ 1 & 0 & 0 & 1 & \\ \hline & 1/6 & 2/6 & 2/6 & 1/6 \end{array}$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h/2, y_n + k_1/2)$$

$$k_3 = hf(x_n + h/2, y_n + k_2/2)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + \frac{k_1}{6} + \frac{2k_2}{6} + \frac{2k_3}{6} + \frac{k_4}{6}$$

Kutta's form

$$\begin{array}{c|cccc} 1/3 & 1/3 & & & \\ 2/3 & -1/3 & 1 & & \\ 1 & 1 & -1 & 1 & \\ \hline & 1/8 & 3/8 & 3/8 & 1/8 \end{array}$$

2nd order RK method

$$k_1 = hf(x_n, y_n)$$
$$k_2 = hf(x_n + \alpha_2 h, y_n + \beta_{21} k_1)$$
$$y_{n+1} = y_n + w_1 k_1 + w_2 k_2$$

$y_{n+1} = y_n + h$ (weighted avg. of slopes on the given interval)

⇒ In general, we find the slope at x_n and several other points in the interval $[x_n, x_{n+1}]$, weighted average these slopes & then multiply by h and add the result to y_n.

A general v-stage RK method is given as

$$y_{n+1} = y_n + w_1 k_1 + w_2 k_2 + \dots w_v k_v, \text{ where}$$
$$k_1 = hf(x_n + \alpha_1 h, y_n + \beta_{11} k_1 + \dots + \beta_{1v} k_v)$$
$$k_2 = hf(x_n + \alpha_2 h, y_n + \beta_{21} k_1 + \dots + \beta_{2v} k_v)$$
$$\vdots$$
$$k_v = hf(x_n + \alpha_v h, y_n + \beta_{v1} k_1 + \dots + \beta_{vv} k_v)$$

where $\alpha_i = \sum_{j=1}^{v} \beta_{ij}$, $i = 1$ to v

We note that the k_i's are defined by a set of v implicit eq., ∴ this is called implicit method.

The method can be written as ⇒

$$\begin{array}{c|cccc} \alpha_1 & \beta_{11} & \beta_{12} & \cdots & \beta_{1v} \\ \alpha_2 & \beta_{21} & \beta_{22} & \cdots & \beta_{2v} \\ \vdots & \vdots & & & \\ \alpha_v & \beta_{v1} & \cdots & \cdots & \beta_{vv} \\ \hline & w_1 & w_2 & \cdots & w_v \end{array}$$

This representation is called Butcher Tableau/ Table.

The matrix $\begin{bmatrix} \beta_{11} & \cdots & \beta_{1v} \\ \beta_{21} & \cdots & \\ \vdots & & \\ \beta_{v1} & \cdots & \beta_{vv} \end{bmatrix}$ is called Runge-Kutta matrix.

For explicit RK method, take $\alpha_1 = 0$ and $\beta_{ij} = 0$ for $i \leq j$ (strictly lower triangular RK matrix)

Explicit RK method can be written as ⇒

$$\begin{array}{c|cccc} 0 & 0 & 0 & \cdots & 0 \\ \alpha_2 & \beta_{21} & 0 & \cdots & 0 \\ \vdots & \vdots & & & \vdots \\ \alpha_v & \beta_{v1} & \cdots & \cdots & 0 \\ \hline & w_1 & w_2 & \cdots & w_v \end{array}$$

Semi-implicit RK method ⇒ $\beta_{ij} = 0$ for $i < j$ in the Butcher tableau. It is also called diagonally implicit method.

v-step semi-implicit RK method is given by ⇒

$$y_{n+1} = y_n + w_1 k_1 + \dots + w_v k_v$$

where
$$k_1 = hf(x_n + \alpha_1 h, y_n + \beta_{11} k_1)$$
$$k_2 = hf(x_n + \alpha_2 h, y_n + \beta_{21} k_1 + \beta_{22} k_2)$$
$$\vdots$$
$$k_v = hf(x_n + \alpha_v h, y_n + \beta_{v1} k_1 \dots + \beta_{vv} k_v)$$

where $\alpha_i = \sum_{j=1}^{i} \beta_{ij}$ $i = 1, \dots, v$

Eg. A 4th order semi-implicit RK method is given by →

$$\begin{array}{c|cccc} 1/2 & 1/2 & 0 & 0 & 0 \\ 2/3 & 0 & 2/3 & 0 & 0 \\ 1/2 & -5/2 & 5/2 & 1/2 & 0 \\ 1/3 & -5/3 & 4/3 & 0 & 2/3 \\ \hline & -1 & \frac{3}{2} & -1 & \frac{3}{2} \end{array}$$

→ Non-zero diagonal

★ α_i → Not negative. Other parameters → Can be -ve.

Eg. Find the solution of DE $y' = 1 + xy$, $y(0) = 1$ by the (classical) 4th order RK method with stepsize = 0.1.

★ Assume explicit method if nothing is specified.

$$k_1 = hf(x_0, y_0)$$
$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$
$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$
$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$y_1 = y_0 + \frac{k_1}{6} + \frac{2k_2}{6} + \frac{2k_3}{6} + \frac{k_4}{6}$$

$x_0 = 0$, $y_0 = 1$ $h = 0.1$ $f(x, y) = 1 + xy$

Now, putting values ⇒

$$k_1 = 0.1(1 + 0 \times 1) = 0.1$$
$$k_2 = 0.1 f(0.05, 1 + 0.05)$$
$$= 0.1 \times (1 + 0.05 \times 1.05)$$
$$= 0.1 \times 1.0525 = 0.10525 = \underline{\underline{0.1053}} \text{ (till 4th place)}$$
$$k_3 = 0.1 f\left(0.05, 1 + \frac{0.1053}{2}\right)$$
$$= 0.1 f(0.05, 1.05265)$$
$$= 0.1 \times (1 + 0.05 \times 1.05265)$$
$$= 0.105263325 = 0.1053$$
$$k_4 = 0.1 \times f(0.1, 1 + 0.1053)$$
$$= 0.1 \times (1 + 0.1 \times 1.1053) = 0.111053 = 0.1111$$

$$y_1 = 1 + \frac{0.1}{6} + \frac{2}{6} \times 0.1053 + \frac{2}{6} \times 0.1053 + \frac{0.1111}{6}$$
$$= 1.1053833\dots$$
$$\boxed{y_1 = 1.1054}$$

★★ Application of numerical method to the test equation $y' = \lambda y$, where λ is a constant ⇒

Exact solution $y(x_{n+1}) = y(x_n + h) = y(x_n) + hy'(x_n) + \frac{h^2}{2!} y''(x_n) + \dots$

Since $y' = \lambda y$, $y'' = \lambda y' = \lambda^2 y$

$y''' = \lambda y'' = \lambda^3 y$ and so on

$$y(x_{n+1}) = y(x_n) + h\lambda y(x_n) + \frac{h^2}{2!}\lambda^2 y(x_n) + \cdots$$

$$= \left(1 + \lambda h + \frac{(\lambda h)^2}{2!} + \frac{(\lambda h)^3}{3!} + \cdots\right) y(x_n)$$

$$\boxed{y(x_{n+1}) = e^{\lambda h}\, y(x_n)}$$

Also, the true solution can be obtained as follows ⇒

$$\frac{dy}{dx} = \lambda y$$

$$\frac{dy}{y} = \lambda\, dx$$

$$\log y = \lambda x + c_1 \longrightarrow \text{Arbitrary constant}$$

$$\Rightarrow y = e^{\lambda x + c_1}$$

$$= e^{c_1} \cdot e^{\lambda x}$$

$$\therefore y = c \cdot e^{\lambda x}, \quad \text{where } c = e^{c_1}$$

$$y(x_{n+1}) = c\cdot e^{\lambda x_{n+1}} = c\cdot e^{\lambda(x_n + h)}$$

$$= c\cdot e^{\lambda x_n}\cdot e^{\lambda h}$$

$$= y(x_n)\cdot e^{\lambda h},$$ which is same as what was obtained by Taylor's expansion.

Solutions by diff. RK methods ⇒

① Euler's method

$$y_{n+1} = y_n + hf(x_n, y_n)$$
$$= y_n + hy_n'$$
$$= y_n + h\lambda y_n$$
$$= (1 + h\lambda)\, y_n$$

Denote $E(\lambda h) = 1 + \lambda h$ so that

$$y_{n+1} = E(\lambda h)\, y_n$$

Note that $E(\lambda h)$ is obtained from the expansion of $e^{\lambda h}$ in powers of h after ignoring terms containing h^2 onwards.

$\therefore E(\lambda h)$ is the 1st order polynomial approx. for $e^{\lambda h}$.

② 2nd order RK method

$$k_1 = hf(x_n, y_n)$$
$$k_2 = hf(x_n + \alpha_2 h,\ y_n + \beta_{21} k_1)$$
$$y_{n+1} = y_n + w_1 k_1 + w_2 k_2$$

$$w_1 + w_2 = 1$$
$$w_2\alpha_2 = 1/2$$
$$\Rightarrow w_2 = \frac{1}{2\alpha_2}$$
$$w_2\beta_{21} = \frac{1}{2}$$
$$\Rightarrow \beta_{21} = \alpha_2 \rightarrow \text{arbitrary}$$
$$w_1 = 1 - w_2 = 1 - \frac{1}{2\alpha_2}$$

$$y_{n+1} = y_n + \left(1 - \frac{1}{2\alpha_2}\right)k_1 + \frac{1}{2\alpha_2}k_2$$

Apply this to the test eq. $y' = \lambda y$

$$f(x, y) = \lambda y$$
$$k_1 = hf(x_n, y_n) = h\lambda y_n$$
$$k_2 = hf(x_n + \alpha_2 h,\ y_n + \alpha_2 k_1)$$
$$= h\lambda(y_n + \alpha_2 k_1)$$
$$= h\lambda(y_n + \alpha_2 h\lambda y_n)$$

$$y_{n+1} = y_n + \left(1 - \frac{1}{2\alpha_2}\right)h\lambda y_n + \frac{1}{2\alpha_2}\cdot\lambda h(y_n + \alpha_2\lambda y_n h)$$

$$= \left[1 + \left(1 - \cancel{\frac{1}{2\alpha_2}}\right)\lambda h + \cancel{\frac{1}{2\alpha_2}\lambda h} + \frac{\cancel{\alpha_2}\lambda^2 h^2}{2\cancel{\alpha_2}}\right] y_n$$

$$y_{n+1} = \left(1 + \lambda h + \frac{\lambda^2 h^2}{2}\right) y_n$$

$y(x_{n+1}) = e^{\lambda h}\, y(x_n)$ is the true solution of the test equation.

Numerically approx. solution → $y_{n+1} = \left(1 + \lambda h + \dfrac{\lambda^2 h^2}{2}\right) y_n$

Putting $E(\lambda h) = 1 + \lambda h + \dfrac{\lambda^2 h^2}{2}$, we get $y_{n+1} = E(\lambda h)\, y_n$

③ 4th order RK method (classical)

$$k_1 = hf(x_n, y_n)$$
$$k_2 = hf(x_n + h/2,\ y_n + k_1/2)$$
$$k_3 = hf(x_n + h/2,\ y_n + k_2/2)$$
$$k_4 = hf(x_n + h,\ y_n + k_3)$$
$$y_{n+1} = y_n + \frac{k_1}{6} + \frac{2k_2}{6} + \frac{2k_3}{6} + \frac{k_4}{6}$$

Apply this method to the test eq. $y' = \lambda y$.

$$k_1 = h\lambda y_n$$
$$k_2 = h\lambda(y_n + k_1/2) = h\lambda(y_n + h\lambda y_n/2)$$
$$= h\lambda y_n + \frac{h^2\lambda^2 y_n}{2}$$

$$k_3 = h\lambda(y_n + k_2/2) = h\lambda\left(y_n + \frac{h\lambda y_n + h^2\lambda^2 y_n/2}{2}\right)$$
$$= h\lambda y_n + \frac{h^2\lambda^2 y_n}{2} + \frac{h^3\lambda^3 y_n}{4}$$

$$k_4 = h\lambda(y_n + k_3) = h\lambda y_n + h^2\lambda^2 y_n + \frac{h^3\lambda^3 y_n}{2} + \frac{h^4\lambda^4 y_n}{4}$$

$$y_{n+1} = y_n + \frac{k_1}{6} + \frac{2k_2}{6} + \frac{2k_3}{6} + \frac{k_4}{6}$$

$$= y_n + \frac{1}{6}\left[h\lambda y_n + 2\left(h\lambda y_n + \frac{h^2\lambda^2 y_n}{2}\right) + 2\left(h\lambda y_n + \frac{h^2\lambda^2 y_n}{2} + \frac{h^3\lambda^3 y_n}{4}\right) + h\lambda y_n + h^2\lambda^2 y_n + \frac{h^3\lambda^3 y_n}{2} + \frac{h^4\lambda^4 y_n}{4}\right]$$

$$= y_n\left(1 + h\lambda + \frac{h^2\lambda^2}{2} + \frac{h^3\lambda^3}{6} + \frac{h^4\lambda^4}{24}\right)$$

$$= y_n\left(1 + h\lambda + \frac{h^2\lambda^2}{2!} + \frac{h^3\lambda^3}{3!} + \frac{h^4\lambda^4}{4!}\right)$$

Now, if $\underbrace{E(\lambda h)}_{4^{th}\text{ order polynomial approx. of } e^{\lambda h}} = 1 + \lambda h + \dfrac{\lambda^2 h^2}{2!} + \dfrac{\lambda^3 h^3}{3!} + \dfrac{\lambda^4 h^4}{4!}$, then we have $y_{n+1} = E(\lambda h)\, y_n$